16. $q_{4,16;0.01} = 5.19$

$W = q_{4,n}\sqrt{\frac{MSE}{n_i}}$

$W = 5.19\sqrt{\frac{471.25}{5}} = 14.91$

$|\bar{X}_6 - \bar{X}_2| = |268.6 - 274.8| = 32.8 > 14.91$

$|\bar{X}_4 - \bar{X}_3| = |268.6 - 271.6| = 17.0 > 14.91$

$|\bar{X}_1 - \bar{X}_2| = 27.4 > 14.91$

$|\bar{X}_1 - \bar{X}_3| = 10.6 < 14.91$

$|\bar{X}_3 - \bar{X}_2| = 16.8 > 14.91$

$|\bar{X}_4 - \bar{X}_1| = 6.4 < 14.91$

At $\alpha = 0.01$, (4 & 2), (4 & 3), (1 & 2) and (3 & 2) are diff. since 2 appears in all the test, it is significantly different

17. a) $\bar{x}_1 = 608$ $\bar{x}_2 = 653$ $\bar{x}_3 = 532$ $\bar{G} = 597.69$

$n_1 = 10$ $n_2 = 10$ $n_3 = 10$

$TSS = \sum (X_{ij} - \bar{G})^2 = 300736.67$

$SST = \sum_{i=1}^{3} (\bar{T}_i - \bar{G})^2 = 74806.67$

$SSE = TSS - SST = 225930$

$MST = \frac{74806.67}{2} = 37403.335$

$MSE = \frac{225930}{27} = 8367.78$

$F = \frac{MST}{MSE} = \frac{37403.335}{8367.78} = 4.47$

$H_0: \mu_1 = \mu_2 = \mu_3$ H_1: at least 1 is diff

$\alpha = 0.05$

RR. $F_{2,27;0.05} \approx 3.35$

since $4.47 > 3.35$, reject H_0

b) $(\bar{x}_1 - \bar{x}_2) \pm q_{3,27;0.05}\sqrt{\frac{MSE}{10}} = (-146.42, 56.42)$

c) $W = q_{3,27}\sqrt{\frac{MSE}{n}} = q_{3,27;0.05} \cdot \sqrt{\frac{8367.78}{10}} = 101.42$

$|\bar{X}_1 - \bar{X}_2| = |608 - 653| = 45 < 101.42$

$|\bar{X}_1 - \bar{X}_3| = |608 - 532| = 76 < 101.42$

$|\bar{X}_2 - \bar{X}_3| = |653 - 532| = 121 > 101.42$

$\therefore$ only physical science & social sciences have significantly different means